EEG-based Anxiety Detection with Feature Selection

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Abstract

Anxiety has become a major public health problem, and accurate indicators of diagnosis are required. Traditional assessments to identify anxiety are quantitative questionnaires, which lack objective criteria that incorporate physiological changes in the body. In recent years, electroencephalography (EEG) has been demonstrated to be a convenient and cost-effective supplementary diagnostic tool. This study presents the methods for selecting the features to extract EEG characteristics. We also employ statistical analyses on feature selection to extract in total 41 important features to objective identify anxiety. The classification results of using these selected and created features were verified through public dataset DASPS. Random Forest has been found to have shown the best performance with a ROC AUC value of 0.93, a recall value of 0.931, and a precision of 94.7%. These findings suggest that EEG signals can potentially serve as reliable indicators to detect anxiety, while providing explainable characteristics that can associate anxiety.

1 Introduction

Anxiety is a kind of mental health problem that is often experienced by people [1]. According to the statistics provided by healthdata.org [2], 1 in 8 people lived with a mental disorder, with anxiety and depression being the most common type. Anxiety becomes a major public health problem worldwide along with depression [3]. Due to the COVID-19 pandemic, the number of people with anxiety and depressive disorders increased greatly in 2020. According to World Health Organization, initial estimates show a 26% and 28% respectively increase in anxiety and depressive disorders in one year [4]. This alarming trend highlights the urgent need for effective strategies to assess and manage anxiety.

Both researchers and physicians have found it difficult to appropriately identify anxiety. Currently, anxiety is generally diagnosed based on recorded symptoms, but there is a significant overlap in the standards for stress, anxiety, and depression, which can lead to misdiagnosis [5]. This highlights the need for an impartial criterion for anxiety detection. Quantitative methods such as questionnaires developed by psychologists are considered reliable for anxiety identification, but they lack objective criteria that involve physiological changes in the body [6]. Although neurologists can generally follow a complex decision tree to systematically evaluate the abnormality of EEG, it is a challenge to visually interpret and especially difficult to perceive signal variations.

Therefore, a reliable automatic clinical EEG diagnosis method without manual intervention is highly needed, which not only reduced the workload of physicians, but also shortened the consultation time for each patient. In the last decades, physiological signals such as electrocardiogram (ECG), photoplethysmography (PPG), electrodermal activity (EDA), and electroencephalography (EEG) have been recognized as potential indicators of physiological reactions associated with

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anxiety [5]. Especially for anxiety diagnosis, EEG has been proved as an economical and convenient auxiliary diagnosis tool. From perspective of diagnosis methods, machine learning models have been proposed to automatic diagnose and detect of the anxiety disorders using EEG signals. However, the majority of current approaches are regarded as ”black boxes” due to the lack of explanations for the decisions made by these models. This can pose difficulties for physicians in understanding and interpreting results.

Thus, the purpose of this paper is to develop a robust anxiety classification model by using a feature selection method and identify the most relevant EEG features that accurately predict anxiety and incorporate them into the classification model, involving pre-processing and analysing the dataset to extract informative features that may be indicative of anxiety. We also investigate and discover new knowledge about EEG patterns associated with anxiety, contributing to the understanding of anxiety diagnosis, with a comprehensive framework for objective anxiety level recognition using EEG-based biomarkers.

The following parts of the paper are organised as following. Section II describes the related work of existing approaches for anxiety detection using EEG. Section III presents our methodology. Section IV shows the experimental results of the method. And the conclusions and future work are presented in section V.

2 Related work

2.1 EEG Patterns Linked to Anxiety

G.Giannakakis et al.[7] studied the measurement of anxiety using electroencephalogram(EEG) signals in reaction to emotional video clips. The asymmetry index value has been found to be increased during a relaxed state and reduced in an anxiety state. A new hardware design for anxiety detection was proposed by A. Arsalan and M. Majid[8], who successfully measured anxiety using high Beta waves. These waves were found to be associated with a high level of anxiety. The alpha and beta bands of EEG were found to be associated with anxiety in the work of L.Richard et al.[9]. The study indicated that a decrease in the alpha power of the right hemisphere and an increase in the beta power were associated with increased anxiety levels. It was also mentioned that the left hemisphere is responsible for processing positive emotions, while the right hemisphere processes negative emotions. A common type of university student's state anxiety called exam anxiety was measured with data extracted from EEG in the work of V.Rajendran et al. [10]. The study found that alpha, beta, and theta waves are connected to anxiety, and physiological indication features (Heart Rate, Cognitive performance attentional resource Index Neural activity, Vigilance index, and Arousal index) calculated from three waves were used to measure the anxiety. It has been observed that in anxiety situations, the alpha and theta waves tend to decrease, while the beta wave becomes more active (increased).

2.2 EEG Classification

Y. Zheng et al.[11] developed a relationship between anxiety and features recovered from EEG and PPG using PCA-kNN, achieving precision of 62. 5%. Anxiety levels were found to be related to wavelet coefficients of the alpha and beta band. In the research of S. Lee et al[12], the driver's anxiety was detected using multiple biosignals, including EEG, PPG, GSR, and pupil size. The EEG signal performed the best with an accuracy of 77.01% using Logistic Regression among the fusion of other signals. A study from L. S. Mokatren[13] investigated another type of state anxiety called social anxiety using EEG signals. Image representation of the EEG data is obtained and Convolutional neural networks (CNN), SVM, and KNN classifiers are trained on the data. With an accuracy of 87%, CNN was deemed to be the top model. C. Chen et al.[14] presented an EEG-neuro feedback method for anxiety identification and classification based on mindfulness recordings and resting-state EEG signals, achieving precision of 87. 18% with a two-label SVM classifier using Power Spectral Density (PSD).

2.3 DASPS dataset and data collection

The use of physiological signals such as EEG, PPG, ECG, and others to measure anxiety in various situations has been the subject of research. However, the number of studies focused on this area is not as plentiful as those examining stress or depression measurements, and the availability of physiological signals data for accurately measuring anxiety is still limited. Despite these challenges, the literature on anxiety measurement is continually expanding, highlighting the need for further exploration in this field. One promising resource for anxiety measurement is the public dataset for evaluating human anxiety with EEG signals, called ”A Database for Anxious States based on Psychological Stimulation (DASPS),” which was published by A. Baghdadi et al.[15]. The data set was collected using a low-cost wireless wearable Emotiv Epoc headset and the authors investigated the effects of trial time, type of feature, combination of features, and number of anxiety levels on EEG data. The study discovered that one

second is the ideal time interval for anxiety stimulation. Later in 2021, the same researchers[16] used this dataset to build a machine to detect anxiety states, achieving 83.50% and 74.60% accuracy for two and four anxiety level classifications, respectively. However, the study did not investigate the effects of EEG electrodes and frequency band selection to optimally classify anxiety into multiple levels. F. Muhammad and S. Al-Ahmadi proposed a new anxiety classification framework using the DASPS dataset by [17]. The study used statistical analysis techniques to select significantly distinct electrodes and identified six channels - AF3, AF4, FC5, FC6, P7, and P8 - for two-level and four-level classification of human anxiety. The best combination of EEG frequency bands was determined using frequency band selection, with the theta and beta bands combination giving the best result. The study used five classification algorithms to classify anxiety levels based on EEG signals, with the random forest classifier achieving the highest accuracy of 94.90% and 92.74% for two and four-level anxiety classification. However, the study focused primarily on improving the performance of the anxiety classification model with the DASPS dataset, rather than investigating which headband's electrodes and frequency bands are affected by anxiety or how frequency bands change based on anxiety level. In summary, the literature on measuring anxiety using physiological signals, particularly EEG signals, is still evolving, and the availability of accurate data for this purpose is limited. However, the DASPS dataset shows promise as a valuable resource for anxiety measurement. Previous studies have used this dataset to develop anxiety classification frameworks and have identified promising combinations of EEG electrodes and frequency bands for anxiety classification. However, the limitations of these studies highlight the need for further exploration in examining the effects of anxiety on specific EEG channels or frequency bands more in-depth. Future research in these areas could contribute to the development of more accurate standards for diagnosing anxiety and ultimately lead to improved outcomes for individuals affected by anxiety disorders.

3 Methodology

3.1 Dataset description

The DASPS database, which contains EEG signals recorded during exposure therapy [15], was utilized in this study. EEG signals were collected from 23 healthy participants (10 males and 13 females) with an average age of 30 years who had no history of mental or physical illness. Before the experiment, the participants signed a consent form and their anxiety levels were measured using the Hamilton Anxiety Rating Scale (HAM-A). HAM-A was used to determine the baseline anxiety levels of the participants and to assess the severity of their anxiety.

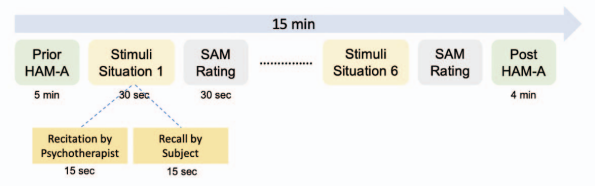

Figure 1. Experiment Process of Data Acquisition [15]

During the experiment, participants were instructed to sit still with their eyes closed while the psychotherapist read aloud a stimuli situation. Participants visualised the situation for the first 15 seconds while the psychotherapist recited it, and then recalled it for the next 15 seconds by themselves. This process was repeated for six different stimuli situations. After each situation, participants rated their feelings on the Self-Assessment Manikin (SAM) scale, which measures valence and arousal. Valence ranges from negative to positive, while arousal ranges from calm to excited. At the end of the experiment, the psychotherapist reassessed the anxiety levels of the participants using a few HAM-A questions. EEG signals were recorded using a wireless headset, Emotiv EPOC, which had 14 channels and 2 mastoids. Electrodes were placed on the scalp at positions AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4, according to the international 10-20 system. EEG data were recorded using Emotiv Epoc software and saved in "Edf" format. The raw data was then converted to "mat" format using a Matlab script for further processing. Each recording took six minutes, with one minute for each stimuli situation, as shown in Fig. 1. The EEG signals were recorded at 128 Hz with an impedance of 7 kilo-ohms. Figures 2 and 3 show two examples of visualizing EEG signals for a participant with severe anxiety and a participant without known anxiety respectively.

3.2 Data Analysis

This study utilised pre-processed EEG signals [15] and six pre-selected channels ('AF3', 'FC5', 'P7', 'P8', 'FC6' and 'AF4') [17] from the DASPS database. Previous research has suggested that Theta, Alpha, and Beta frequencies may be relevant to anxiety in terms of the frequency domain, and thus, these frequencies were the focus of this study.

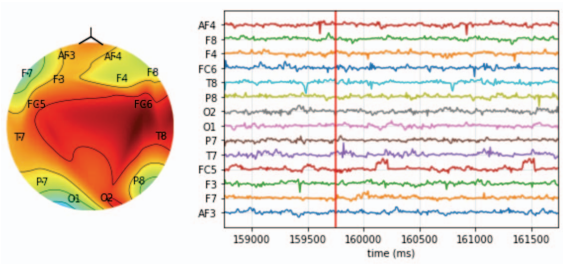

Figure 2. Visualization of EEG signal for a participant known with severe anxiety. (left) the heatmap of scalp at the time marked by the vertical line on the right plot; (right) EEG signals in the time domain. y axis is mV (Millivolts) of the named electrodes.

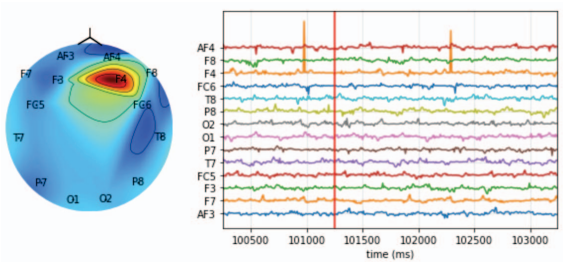

Figure 3. Visualization of EEG signal for a participant without known anxiety. (left) the heatmap of scalp at the time marked by the vertical line on the right plot; (right) EEG signals in the time domain. y axis is mV (Millivolts) of the named electrodes.

3.2.1 Data Labelling

The study used the Self-Assessment Manikin (SAM) test to label the EEG data with anxiety states instead of the Hamilton Anxiety Rating Scale (HAM) test. This decision was made because the last HAM result obtained could not represent the six situations the participant experienced. The labelling process was carried out using the following criteria [18]: data points with valence scores greater than or equal to 3 and arousal scores less than or equal to 5 were labelled as "Light Anxiety", while data points with valence scores less than or equal to 2 and arousal scores less than or equal to 3 were also labelled as "Light Anxiety". On the other hand, data points with valence scores greater than or equal to 3 and arousal scores greater than or equal to 6 were labelled as "Severe Anxiety", while data points with valence scores less than or equal to 2 and arousal scores greater than or equal to 4 were also labelled as "Severe Anxiety".

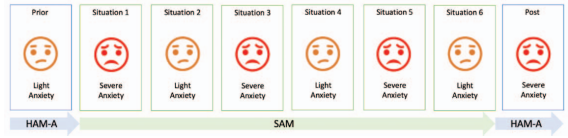

Figure 4. Example of Subject's Anxiety Level Assessment Result

3.2.2 Imbalanced Data

The number of observations in the Light Anxiety group was significantly smaller than in the Severe Anxiety group, indicating imbalanced data. To address this issue, we used SMOTE-EEN (Synthetic Minority Oversampling Technique with Edited Nearest Neighbours) to balance the number of observations in each group. SMOTE-EEN is a technique that generates synthetic samples of the minority class (in this case, the Light Anxiety group) using k-nearest neighbours and then removes noisy and borderline samples.

3.2.3 Statistical Analysis

For frequency domain data, the study used a permutation test to compare the Power Spectral Density (PSD) of the three waves, theta, alpha, and beta, in the six channels between participants with severe anxiety and light anxiety. This method was selected because the underlying distribution of the data is not normally distributed, which can pose a challenge when using traditional parametric statistical tests such as t-tests or ANOVA. The participant labels were randomly permuted and the PSD difference between the two groups was calculated. The difference observed in PSD was compared to the null distribution to see if it was statistically significant. This process was performed numerous times to create a null distribution of differences in PSD. By using a permutation test, the study was able to test for group differences in PSD while controlling for potential confounding variables and avoiding assumptions about the distribution of the data.

3.3 Feature Extraction and Feature Selection

3.3.1 Time Domain Feature

The study utilized root mean square (RMS), peak-to-peak value, and median as time domain features of EEG signals to investigate differences between emotional states. These features were chosen because they are commonly used to quantify signal amplitude and variability, which can reflect changes in brain activity

related to emotional processing. RMS value's mathematical equation is as follows:

$$V_{\text{RMS}} = \sqrt{\frac{1}{T}\int_0^T V(t)^2\,dt} \tag{1}$$

Peak-to-peak value is the difference between the highest and the lowest voltage values.

3.3.2 Frequency Domain Feature

Frequency domain features i.e. band power are popular features in emotion recognition with a signal. The band power of EEG signal is defined as follows:

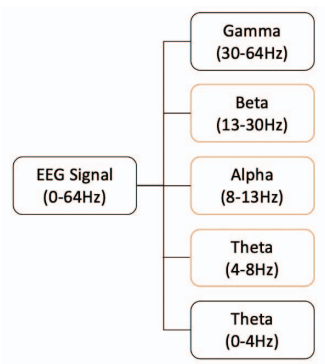

Figure 5. Band Power of EEG Signal

Frequency features are extracted with MNE Python package used to explore, visualize, and analyze human neurophysiological data such as MEG, EEG, and others. Welch's method with an overlap window of 50% is used to calculate the power spectral density (PSD) for each power band. With PSD, mean power (MP), rational asymmetry (RASM), and asymmetry index (AI) features are calculated. RASM is calculated as follows [19]:

$$\frac{P_i^{\text{Left}}}{P_i^{\text{Right}}} = \text{RASM}_i \tag{2}$$

AI is calculated as follows [20]:

$$\frac{P_{\text{Right}} - P_{\text{Left}}}{P_{\text{Right}} + P_{\text{Left}}} \times 100 = \text{AI} \tag{3}$$

AI has a broader concept compared to RASM. A positive value of AI indicates greater activity in the right hemisphere of the brain, while a negative value indicates greater activity in the left hemisphere. The selection of frequency domain features was based on the results of the statistical analysis conducted in the previous section. Features with a p-value less than 0.05 were considered statistically significant and were selected for classification.

3.4 Classification

The classification part of the project involves training and testing several machine learning classifiers on the features extracted from the EEG signals. In this study, K-Nearest Neighbor (KNN), Multi-Layer Perceptron (MLP), Random Forest (RF), eXtreme Gradient Boosting (XGB), and Voting classifiers were used. KNN is a non-parametric method that classifies new data points by comparing them to their k-nearest neighbours in the training set. MLP is a feedforward neural network that learns its weights and biases using the back-propagation technique. RF is an ensemble learning method that creates numerous decision trees and combines their predictions to improve accuracy and reduce over-fitting. XGB is a gradient-boosting approach that creates a model by iteratively adding decision trees to minimize the loss function. For hyper-parameter tuning, Grid Search with Cross-Validation was done. Grid Search is an exhaustive search algorithm that tests all possible combinations of hyper-parameters to find the best-performing model. Cross-validation was used to ensure the robustness of the models. The models' performances were evaluated with cross-validation and confusion matrix. During model training, cross-validation was performed with 5 folds for each classifier. The classifiers were trained on the over-sampled data and evaluated on the testing set within each fold. It prevented over-fitting and ensured the robustness of the model. In the testing stage, accuracy, precision, recall, F1-score, and confusion matrix were evaluated for a detailed understanding of the strengths and weaknesses of each classifier. For the top 2 best-performing models, Area Under the Receiver Operating Characteristic curve (AUC-ROC) metrics were also evaluated in addition to accuracy, precision, recall, F1-score, and confusion matrix.

In the study, the priority of factors for final model evaluation was determined to be ROC AUC, recall, and accuracy, in that order. This is because false negatives, which occur when a severe case of anxiety is misclassified as a light case, can have significant consequences and are considered very costly. Therefore, the focus is on minimizing false negatives, which is the number of instances that are actually positive but predicted as negative. In such cases, ROC AUC and recall values can provide valuable insights into the performance of a classification model and help identify the best-performing model.

4 Experiment Results

4.1 Overview

To obtain relevant information from the EEG signals, the recording for each subject was segmented for a 1-

second duration, which has been previously identified as optimal for anxiety stimulation based on previous research [15]. Thus, there were EEG signal records of 180 seconds which are 180 instances for each subject. With 23 subjects, this resulted in a dataset of 4140 instances for analysis. The data labelling results using the SAM score indicate that out of the total number of cases analyzed, 2580 instances were classified as having severe anxiety, while 1560 were classified as having light anxiety. Fig 6 shows the distribution of anxiety Levels of this dataset in Valence-Arousal Space.

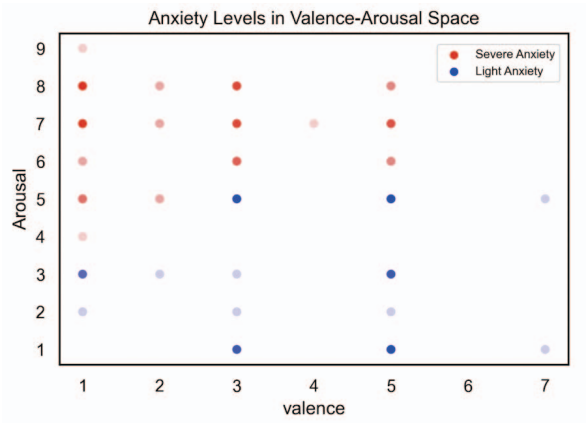

Figure 6. Anxiety Levels of Data in Valence-Arousal Space

The imbalanced dataset was addressed with the SMOTE-EEN technique. The result of applying it to the dataset revealed that the number of instances classified as having light anxiety increased from 1560 to 1439, while the number of instances classified as having severe anxiety decreased from 2580 to 1404.

4.2 Statistical Analysis

A total of 18 features, representing 3 brain waves (theta, alpha and beta) across 6 channels('AF3', 'FC5', 'P7', 'P8', 'FC6', and 'AF4'), were examined. The statistical analysis of these features revealed that 9 features had a p-value less than 0.05, indicating their statistical significance in distinguishing between severe and light anxiety. Specifically, the analysis revealed that the right hemisphere of the brain was significantly related to anxiety levels. It is because the result showed that among 9 statistically significant features, 6 features were obtained from the right hemisphere. Additionally, since 5 features of them are alpha waves, it indicated that alpha waves were found to be the most related brain wave to anxiety level and showed its potential as an indicator of anxiety.

4.3 Feature Extraction and Feature Selection

Time domain features including root mean square (RMS), peak-to-peak value, and median were computed from the EEG signals collected from 6 channels, resulting in a total of 18 time domain features. Meanwhile, frequency domain features were extracted from the theta, alpha, and beta waves in the EEG signals. Specifically, 18 mean power (MP) features were computed for each frequency band (theta, alpha, and beta) and for each of the 6 channels. Additionally, 6 Rational Asymmetry (RASM) features and 6 Asymmetry Index (AI) features were calculated for each wave and each channel. In total, 36 frequency domain features were extracted from the EEG signal. Using the results of statistical analysis, a total of 9 MP features were selected as being statistically significant for the classification. Specifically, the theta waves of AF3 and P8, as well as the alpha waves of AF3, P7, P8, FC6, and AF4, and the beta waves of P8 and AF4 showed significant differences in power between the two anxiety groups. For RASM and AI features, features of the theta waves of AF and P, the alpha waves of AF, FC, and P, and the beta waves of AF and P were selected based on the p-value. As a result, a total of 41 features were selected for classification, including 23 features from the frequency domain and 18 features from the time domain.

4.4 Classification

As a result of performing hyper-parameter tuning, the optimal hyper-parameters for each classifier were determined. The K-Nearest Neighbor (KNN) classifier was found to perform best with a number of neighbours set to 3. The Multi-Layer Perceptron (MLP) classifier was optimized by setting the number of hidden layers to 6 and using a batch size of 32. For the Random Forest (RF) classifier, the optimal number of trees was set to 200 with a maximum depth of each tree set to 20. The eXtreme Gradient Boosting (XGB) classifier was tuned to use a learning rate of 0.1 with 100 trees and a maximum depth of each tree set to 3. Finally, the Voting classifier was optimized by combining KNN and RF with a soft voting method.

As listed in Table 1, the trained classifiers were evaluated on their accuracy, recall, precision, and F1 score. KNN achieved an accuracy of 92.1%, a recall of 0.909, a precision of 0.929, and an F1-score of 0.919. MLP obtained 83.6% for accuracy, 0.901 for recall value, 0.791 for precision value, and 0.843 for F1-score. RF achieved an accuracy of 93.1%, a recall of 0.947, a precision of 0.914, and an F1-score of 0.930. XGB achieved 0.925 for accuracy, 0.928 for a recall value, 0.919 for a pre-

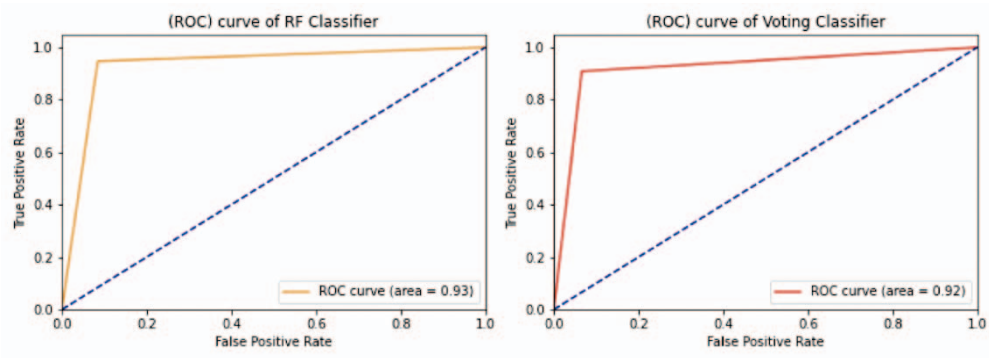

Figure 7. (ROC) curve of RF and Voting Classifiers

Table 1. Classifier Comparison: Accuracy, Recall, Precision, and F1 Scores

Classifier	Accuracy (%)	Recall	Precision	F1 Score
KNN	92.1	0.909	0.929	0.919
MLP	83.6	0.901	0.791	0.843
RF	93.1	0.947	0.914	0.930
XGB	92.5	0.928	0.919	0.923
Voting	94.4	0.930	0.953	0.942

cision value, and 0.923 for F1-score. Finally, Voting achieved an accuracy of 94.4%, a recall of 0.930, a precision of 0.953, and an F1-score of 0.942. Based on the results, it can be seen that the Voting classifier achieved the highest accuracy of 94.4%, followed by RF with 93.1%.

The performance of the two top-performing models, the Voting and RF classifiers, were evaluated using ROC AUC as well as other metrics such as accuracy and recall. As in Fig. 7, the ROC AUC evaluation showed that the RF classifier outperformed the Voting classifier with a higher ROC AUC value of 0.93 compared to 0.92. Furthermore, it was found that the RF classifier had a higher recall value of 0.947 compared to the Voting classifier. As a result, based on the evaluation criteria discussed in the Methodology section, it can be concluded that the RF classifier is the most suitable classifier for anxiety classification.

5 Conclusions

In this study, EEG signals were explored as an objective means of diagnosing anxiety, using a public data set called "A Database of Anxious States Based on Psychological Stimulation (DASPS)". EEG features related to Anxiety levels were found out and Anxiety Classification Model with high performance was built. Data labelling was done with Self-Assessment Manikin (SAM) scale results and obtained Severe Anxiety and Light Anxiety labels. EEG signals from six channels, including AF3, AF4, FC5, FC6, P7, and P8, were extracted to obtain characteristics in the time domain and in the frequency domain, and the selection of features from statistical analysis was used to extract a total of 41 characteristics. The result of the statistical analysis showed that activation of the right hemisphere of the brain and alpha waves are related to anxiety levels. Anxiety classification was performed using five different classifiers, which are KNN, MLP, RF, XGB and Voting. According to the model evaluation criteria, the Random Forest classifier with a ROC AUC value of 0.93, a recall value of 0.931, and an accuracy of 94.7% was considered the most suitable model for anxiety classification. The results of the project offer new evidence in favour of the use of EEG data for anxiety assessment and raise the possibility of building a more precise and individualised diagnosis model for anxiety. Future work could explore the use of additional EEG features and incorporate other physiological signals to enhance the performance of the anxiety classification model.

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